

**ON FORMAL VERIFICATION TECHNIQUES FOR MASKED
CIRCUITS: APPROACHES AND THEORETICAL
UNDERPINNINGS**

by
KAĞAN GÜLSÜM

Submitted to the Graduate School of Engineering and Natural Sciences
in partial fulfilment of
the requirements for the degree of Master of Science
in Computer Science and Engineering

Sabancı University
July 2026

**ON FORMAL VERIFICATION TECHNIQUES FOR MASKED
CIRCUITS: APPROACHES AND THEORETICAL
UNDERPINNINGS**

Approved by:

Asst. Prof. SÜHA ORHUN MUTLUERGİL
(Thesis Supervisor)

Prof. ERKAY SAVAŞ

Asst. Prof. KLAUS FREIHERR VON GLEISSENTHAL

Date of Approval: July 22, 2026

KAĞAN GÜLSÜM 2026 ©

All Rights Reserved

ABSTRACT

ON FORMAL VERIFICATION TECHNIQUES FOR MASKED CIRCUITS: APPROACHES AND THEORETICAL UNDERPINNINGS

KAĞAN GÜLSÜM

Computer Science and Engineering, M.Sc. Thesis, July 2026

Thesis Supervisor: Asst. Prof. SÜHA ORHUN MUTLUERGİL

Keywords: keyword1, keyword2, keyword3, keyword4

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante. Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc, Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. et magnis dis parturient montes, nascetur ridiculus mus.

ÖZET

MASKELENMİŞ DEVRELER İÇİN FORMAL DOĞRULAMA TEKNİKLERİ ÜZERİNE: YAKLAŞIMLAR VE KURAMSAL TEMELLER

KAĞAN GÜLSÜM

Bilgisayar Bilimi ve Mühendisliği, Yüksek Lisans Tezi, Temmuz 2026

Tez Danışmanı: Dr. Öğr. Üyesi SÜHA ORHUN MUTLUERGİL

Anahtar Kelimeler: kelime1, kelime2, kelime3, kelime4

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu. In enim justo, rhoncus ut, imperdiet a, venenatis vitae, justo. Nullam dictum felis eu pede mollis pretium. Integer tincidunt. Cras dapibus. Vivamus elementum semper nisi. Aenean vulputate eleifend tellus. Aenean leo ligula, porttitor eu, consequat vitae, eleifend ac, enim. Aliquam lorem ante, dapibus in, viverra quis, feugiat a, tellus. Phasellus viverra nulla ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maecenas nec odio et ante tincidunt tempus. Donec vitae sapien ut libero venenatis faucibus. Nullam quis ante. Etiam sit amet orci eget eros faucibus tincidunt. Duis leo. Sed fringilla mauris sit amet nibh. Donec sodales sagittis magna. Sed consequat, leo eget bibendum sodales, augue velit cursus nunc, Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. et magnis dis parturient montes, nascetur ridiculus mus.

ACKNOWLEDGEMENTS

I would like to thank my advisor and family.

Dedicated to Okan Buruk...

TABLE OF CONTENTS

ABSTRACT	iv
ÖZET	v
LIST OF TABLES	xiii
LIST OF FIGURES	xiv
LIST OF SYMBOLS	xv
LIST OF ABBREVIATIONS	xvi
1. INTRODUCTION	1
1.1. Side-Channel Attacks and the Masking Countermeasure.....	1
1.2. Verifying Masked Implementations	1
1.3. Scope and Threat Model	1
1.4. Contributions.....	1
1.5. Thesis Outline	1
2. PRELIMINARIES	2
2.1. Notation and Mathematical Background	3
2.1.1. The Field $\mathbb{F}_2$ and Multilinear Boolean Polynomials.....	3
2.1.2. Probability Distributions and Statistical Independence.....	3
2.1.3. Boolean Circuits and Gates	3
2.2. Secret Sharing and Masking.....	3
2.2.1. Boolean Masking	3
2.3. The ISW Probing Model	3
2.3.1. t -Threshold Probing and Observation Sets.....	3
2.3.2. d -Probing Security	3
2.3.3. Probabilistic Non-Interference	3
2.3.4. Threshold and Strong Non-Interference	3

2.4. Optimistic Sampling.....	3
2.5. Computational Hardness Preview	3
3. RELATED WORK	4
3.1. Language-Based Verification	5
3.1.1. Verified Proofs of Higher-Order Masking	5
3.1.2. Large Sets and the Exploration Algorithm.....	5
3.1.3. Witness Replay and Derivation Reuse	5
3.1.4. MaskVerif With Physical Defaults	5
3.2. Fourier-Analytic Verification	5
3.2.1. The Xiao--Massey Criterion and Approximations.....	5
3.2.2. SAT-Based Realization and Its Cost	5
3.3. Other Approaches	5
3.3.1. Model Counting and SMT	5
3.3.2. Direct Linear-Algebraic Methods	5
3.3.3. Compositional Reasoning and Relabeling	5
3.4. Position of This Work	5
4. COUPLINGS AND PROBING SECURITY	6
4.1. The Counting Problem Is #P-Complete	7
4.1.1. From Independence to Distribution Counting	7
4.1.2. #P-Completeness and Its Consequences for Verification.....	7
4.2. Couplings of Probabilistic Programs	7
4.2.1. Coupling Two Runs Under Different Secrets	7
4.2.2. Coupling Functions: Randoms to Randoms.....	7
4.3. Probing Security via Couplings	7
4.3.1. d -Probing Security Implies a (Family of) Coupling Functions ..	7
4.3.2. Identity-On-Leakage Couplings and Secret-Indistinguishability	7
4.4. Optimistic Sampling as Coupling Construction	7
4.4.1. Rewrites as Bijections on the Random Space	7
4.4.2. The Derivation as an Explicit Coupling Witness	7
4.4.3. Why a Successful Rewrite Yields a Valid Coupling	7
5. DESIGN OF XXX	8
5.1. Pipeline Overview	9

5.2. Expression Representation	9
5.2.1. A Hash-Consed n -Ary Circuit DAG	9
5.2.2. Canonicalization: Flattening and Cancellation	9
5.2.3. Multiplicative Factoring.....	9
5.2.3.1. The Factoring Transformation.....	9
5.2.3.2. Enlarging the Class of Verifiable Circuits	9
5.2.3.3. Cost Analysis and the Factoring-Speed Tension	9
5.3. Component I: The Rewrite Engine	9
5.3.1. The Simple Rule: Reference-Counted Optimistic Sampling.....	9
5.3.2. First Fallback: General Substitution	9
5.3.3. Relation to the Coupling-Construction View	9
5.3.4. Last Fallback: Counting	9
5.4. Component II: Traversing the Subset Space.....	9
5.4.1. The Space-Splitting Worklist Representation	9
5.4.2. Large-Set Pruning and Closure-Based Splitting.....	9
5.4.3. Completeness via Vandermonde’s Identity	9
5.5. Component III: Extending Probe Sets	9
5.5.1. Closures: Structural Deduction of Free Wires	9
5.5.2. Coupling-Driven Extension	9
6. IMPLEMENTATION OF XXX	10
7. EVALUATION.....	11
7.1. Benchmarks	11
7.2. Methodology and Metrics	11
7.3. Results	11
7.3.1. Verification Outcomes and Case Studies	11
7.3.2. Redundancy in the Search	11
7.3.3. The Cost and Benefit of Extending Probe Sets	11
7.4. Discussion	11
8. FUTURE WORK	12
8.1. Arithmetic Circuits.....	12
8.2. Symmetry-Orbit Space Exploration	12
8.3. Randomized Fast-Reject for the Engine	12

8.4. Scaling and Compositional Verification	12
8.5. Glitches and Practical Aspects	12
9. CONCLUSION	13
APPENDIX.....	14

LIST OF TABLES

LIST OF FIGURES

LIST OF SYMBOLS

ϕ Coupling function.

LIST OF ABBREVIATIONS

ISW Ishai, Sahai, Wagner.

1. INTRODUCTION

1.1 Side-Channel Attacks and the Masking Countermeasure

1.2 Verifying Masked Implementations

1.3 Scope and Threat Model

1.4 Contributions

1.5 Thesis Outline

2. PRELIMINARIES

2.1 Notation and Mathematical Background

2.1.1 The Field $\mathbb{F}_2$ and Multilinear Boolean Polynomials

2.1.2 Probability Distributions and Statistical Independence

2.1.3 Boolean Circuits and Gates

2.2 Secret Sharing and Masking

2.2.1 Boolean Masking

2.3 The ISW Probing Model

2.3.1 t -Threshold Probing and Observation Sets

2.3.2 d -Probing Security

2.3.3 Probabilistic Non-Interference

2.3.4 Threshold and Strong Non-Interference

3. RELATED WORK

3.1 Language-Based Verification

3.1.1 Verified Proofs of Higher-Order Masking

3.1.2 Large Sets and the Exploration Algorithm

3.1.3 Witness Replay and Derivation Reuse

3.1.4 MaskVerif With Physical Defaults

3.2 Fourier-Analytic Verification

3.2.1 The Xiao--Massey Criterion and Approximations

3.2.2 SAT-Based Realization and Its Cost

3.3 Other Approaches

3.3.1 Model Counting and SMT

3.3.2 Direct Linear-Algebraic Methods

4. COUPLINGS AND PROBING SECURITY

4.1 The Counting Problem Is #P-Complete

4.1.1 From Independence to Distribution Counting

4.1.2 #P-Completeness and Its Consequences for Verification

4.2 Couplings of Probabilistic Programs

4.2.1 Coupling Two Runs Under Different Secrets

4.2.2 Coupling Functions: Randoms to Randoms

4.3 Probing Security via Couplings

4.3.1 d -Probing Security Implies a (Family of) Coupling Functions

4.3.2 Identity-On-Leakage Couplings and Secret-Indistinguishability

4.4 Optimistic Sampling as Coupling Construction

4.4.1 Rewrites as Bijections on the Random Space

5. DESIGN OF XXX

5.1 Pipeline Overview

5.2 Expression Representation

5.2.1 A Hash-Consed n -Ary Circuit DAG

5.2.2 Canonicalization: Flattening and Cancellation

5.2.3 Multiplicative Factoring

5.2.3.1 The Factoring Transformation

5.2.3.2 Enlarging the Class of Verifiable Circuits

5.2.3.3 Cost Analysis and the Factoring–Speed Tension

5.3 Component I: The Rewrite Engine

5.3.1 The Simple Rule: Reference-Counted Optimistic Sampling

5.3.2 First Fallback: General Substitution

6. IMPLEMENTATION OF XXX

7. EVALUATION

7.1 Benchmarks

7.2 Methodology and Metrics

7.3 Results

7.3.1 Verification Outcomes and Case Studies

7.3.2 Redundancy in the Search

7.3.3 The Cost and Benefit of Extending Probe Sets

7.4 Discussion

8. FUTURE WORK

8.1 Arithmetic Circuits

8.2 Symmetry-Orbit Space Exploration

8.3 Randomized Fast-Reject for the Engine

8.4 Scaling and Compositional Verification

8.5 Glitches and Practical Aspects

9. CONCLUSION

APPENDIX